Kubernetes Persistent Volume Mount Issue SOP

Objective

Restore persistent storage access when Kubernetes pods cannot mount or bind persistent volumes and claims.

Scope

Applicable to Kubernetes workloads, platform operations, DevOps teams, SRE teams, application teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

1. Access to Kubernetes cluster using kubectl and the correct namespace context.
2. Access to application, pod, deployment, service, ingress, and node logs where applicable.
3. Access to monitoring dashboards such as Grafana, Prometheus, cloud monitoring, or container insights.
4. Permission to inspect manifests, secrets, configmaps, RBAC bindings, PVCs, and rollout history.
5. Escalation contact for DevOps, platform, infrastructure, network, and application teams.

Incident Metadata

Field	Value
Ticket ID	T045
Issue	Kubernetes persistent volume issue
Category	Storage
Service	Kubernetes
Severity	Medium
Status	Resolved
Description	Volume not mounting
Expected Resolution	Check PV and PVC binding

Procedure

1. Confirm PVC name, namespace, storage class, pod name, and mount error message.
2. Check PVC phase, requested size, access mode, and bound PV.
3. Describe the PVC and pod to review provisioning, attach, or mount events.
4. Validate storage class availability and dynamic provisioner status.
5. Check access mode compatibility such as ReadWriteOnce, ReadWriteMany, or ReadOnlyMany.
6. Verify node attach limits, cloud disk state, and filesystem permissions.
7. Fix PVC, storage class, access mode, provisioner, or mount path configuration.
8. Restart affected pods and confirm volume is mounted successfully.

Common Commands

```
kubectl config current-context
kubectl get pods -n <namespace> -o wide
kubectl describe pod <pod-name> -n <namespace>
kubectl logs <pod-name> -n <namespace> --tail=100
kubectl get events -n <namespace> --sort-by=.lastTimestamp
kubectl get deploy,svc,ingress,pvc -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Volume not mounting	PVC not bound	Check PV/PVC binding and storage class
Attach failed	Cloud disk or node limit	Check disk attachment and node capacity
Permission denied	Filesystem permissions	Fix security context or volume permissions
Wrong access mode	Multiple pods using RWO volume	Use compatible access mode or scheduling

Best Practices

1. Monitor PVC pending status.
2. Document storage class and access mode rules.
3. Avoid deleting PVCs without data-impact review.
4. Use backups for critical persistent data.

References

Kubernetes Docs **|** Internal Tickets CSV **|** Cluster Events **|** Monitoring Dashboards **|** Application Logs **|** Platform Runbook